

Nijat Taghizade

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Objective

Enthusiastic first year Electrical Engineering student with prior experience in programming in both high-level and low-level programming languages, machine learning systems, 3D Design software, microcontroller architecture, and familiarity with laboratory environments looking to contribute as an Undergraduate Researcher in the ORS program.

Education

Georgia Institute of Technology | Atlanta, GA

Bachelor of Science in Electrical Engineering, GPA: 4.00

August 2025 – Present

Expected Graduation, May 2029

Skills

Programming: Python, C++, Java, SQL

Hardware: Raspberry Pi, FPGAs (Verilog/SystemVerilog), PCB Design (Altium), Oscilloscope

Software: MATLAB, Fusion 360, AutoCAD, React, VSCode, Jupyter

Communication: Design proposals, technical reports, instruction manuals, presentations, analytical research papers

Languages: English (Fluent), German (Fluent), Russian (Conversational), Turkish (Conversational)

Experience

Georgia Institute of Technology | Atlanta, Georgia

September 2025 – Present

Undergraduate Research / Biomedical Microsystems Laboratory

- Designed a motor-driven system to automate cell mixing inside a bioreactor, replacing manual rotation and tilt routines with programmable motion profiles.
- Prototyped firmware for timed rotations and tilt sequences for operator time reduction and boosted efficiency.

Cambridge University | Cambridge, UK

February 2025 – May 2025

Research Student / Computer Science and Bioinformatics

- Conducted research in machine learning for fault detection within three-phase electrical transmission systems, achieved 99% accuracy on integrated models.
- Integrated transformer and line parameters as model features to improve detection precision, the model demonstrated annual energy savings of approximately 500 megajoules and an ROC AUC of 0.99.

University of North Georgia | Remote

September 2025 – Present

Remote Researcher (Independent, Faculty Supervision)

- Conducted independent research on adapting autoregressive language models to represent and generate crystal structures using CIF-style sequences derived from the CrystaLLM dataset.
- Fine-tuned LLaMA (3-8b) models using LoRA (Low-Rank Adaptation) to reduce trainable parameters and enable efficient fine-tuning.

Projects

First Robotics Team | Captain of Electronics and Design

March 2023- March 2025

- Led electronics, overseeing implementations of power trees, protection (fuses/TVS), grounding and cable management for a 50 kg robot, achieved >99% event uptime.
- Implemented CAN bus circuit system and diagnostics, wrote Java telemetry for live motor currents, cutting fault recovery time from 2 min to 20s.

PathFindr |(AI ATL Hackathon Winner)

October 2025 - Present

- iOS application for low and no vision users that provided step-by-step indoor navigation and concise outdoor scene descriptions using LiDAR-based depth maps.
- Built a multi-agent pipeline with Google's ADK (Agent Development Kit), creating specialized gemini-powered agents (hazard detection, spatial narration and dialogue) with a hazard-first prioritization system.

Relevant Coursework

Calculus 1-3, Linear Algebra: Linear approximation/Taylor methods, multiple integration, and vector calculus (Green/Gauss/Stokes), matrices and linear systems, eigenvalues/eigenvectors

Signal Processing: Discrete-time signals and linear systems: sampling, filtering, frequency response, DFT, and Z-transform, with MATLAB-based lab work

Digital System Design: Logic gates/switches, Boolean algebra, number systems/arithmetic, storage elements, and intro datapath/memory/ISA concepts

Physics: Calculus-based mechanics with lab; motion and forces, work/energy, momentum, rotation/statics, plus oscillations and waves topics